

Test 1

Assignment

- G signifies students that take this class for **Graduate Credit**; they must do all the items
- UG students are **not** required to do the G work, but they may do it for bonus points
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

NOTES:

- Show your work to obtain partial credit if appropriate
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement
https://en.wikipedia.org/wiki/International_System_of_Units
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga...)
https://en.wikipedia.org/wiki/Metric_prefix
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit
- Normalized plots are plots with the y-axis scaled between -1 and $+1$. A plotted variable is normalized by dividing it by its maximum absolute value

Notations and abbreviations:

- BVP = boundary value problem
- EOM = equation of motion
- FBD = free body diagram
- FRF = frequency response function
- IVP = initial value problem
- LHS = left hand side
- NLM = Newton's law of motion
- ODE = ordinary differential equation
- PDE = partial differential equation
- RHS = right hand side

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A. Pendulum

A.1 ANALYZE PENDULUM

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A.2 CALCULATE PENDULUM RESPONSE TO INITIAL DISPLACEMENT

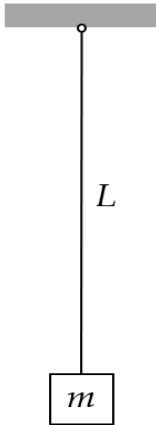


Figure 1: pendulum of length L and mass m

Take $g=9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L_{\text{nominal}}=500\text{mm}$ and mass $m=45\text{kg}$ as shown in Figure 1. When manufactured in series production, the pendulum length values are not exact due to manufacturing tolerance. The mm length values for ten pendulum sets are:

$L = [502.013, 499.337, 500.472, 502.949, 502.216, 496.635, 500.416, 498.255, 498.350, 499.358] \text{ mm}$

Do the following:

(1) For the ten pendulum sets, calculate

- rad/sec natural frequency ω_n
- Hz natural frequency f_n
- period of oscillation τ
- display in table format L, ω_n, f_n, τ

(2) Mean values: calculate and display the mean values of ω_n, f_n, τ

- (3) Initial value problem (IVP): assume initial displacement $u_0=4\text{mm}$ and zero initial velocity. Use the mean values of ω_n, f_n, τ to do the following:
- (a) Calculate and display the position of the mass $u(t_1)$ after $t_1=18\text{ sec}$
 - (b) Calculate and display the duration $t_{N_{osc}}$ of N_{osc} oscillations and the corresponding position of the mass $u_{N_{osc}}$ for $N_{osc}=15$
 - (c) Plot the response of the system for N_{osc} oscillations and use a datatip on the plot to find the value $u(t_1)$ at t_1 on the plot
- (4) Discuss your results

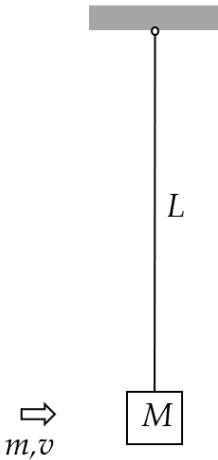
A.3 CALCULATE PENDULUM RESPONSE TO IMPACT

Figure 2: impact excitation of a pendulum system

Take $g=9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L=500 \text{ mm}$ and mass $M=45\text{kg}$ as shown in Figure 2. The pendulum is initially at rest hanging vertically. A small projectile of mass $m=11 \text{ grams}$ and velocity $v=850 \text{ m/s}$ hits the mass M and remains embedded into it. This impact imparts an initial velocity to the system that starts to oscillate. Do the following:

- (1) display input data
- (2) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (3) Calculate the rad/sec natural frequency ω_n and period of oscillation τ of the pendulum with projectile embedded into it
- (4) Calculate the response $u(t_1)$ after $t_1=18 \text{ sec}$
- (5) Plot the response of the system for $N_{osc}=15$ oscillations and find on the plot:
 - (a) the largest response
 - (b) the response value calculated at item (4)
- (6) Discuss your results

B. Spring-mass-damper vibrating systems

B.1 ANALYZE SPRING-MASS VIBRATING SYSTEM

NOTHING HERE

B.2 ANALYZE SPRING-MASS-DAMPER VIBRATING SYSTEM

NOTHING HERE

B.3 CALCULATE SPRING-MASS-DAMPER RESPONSE TO INITIAL DISPLACEMENT

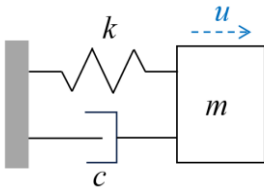


Figure 3: spring-mass-damper vibrating system

Consider a spring-mass vibrating system with $m=5.3$ kg, $k=530$ N/m, $c=22$ N/(m/s) as shown in Figure 3. The vibrating system is given an initial displacement $u_0=4$ mm with zero initial velocity. Do the following:

- (1) Display input data
- (2) Calculate the natural frequency in rad/sec ω_n and in Hz f_n
- (3) Calculate the critical damping c_{cr}
- (4) Calculate the damping ratio ζ
- (5) Calculate the damped frequency in rad/sec ω_d and in Hz f_d
- (6) Calculate the damped period of oscillation τ_d
- (7) Calculate the characteristic-equation roots s_1 , s_2 and the constants C_1 , C_2
- (8) Calculate the constants A , B and then the constants C , φ
- (9) Calculate the duration t_1 of one oscillation and the corresponding position $u_1=u(t_1)$
- (10) Calculate the value $u_{0.02}$ representing 2% of the initial displacement
- (11) Plot the response of the system for $N_{osc}=12$ oscillations and use datatips on the plot to find the following:
 - (a) the largest response
 - (b) the response value calculated at item (9)
 - (c) the time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (10)
- (12) Discuss your results

B.4 CALCULATE TRUCK SUSPENSION RESPONSE

Consider a pickup truck wheel suspension consisting of a leaf spring and an oil damper. The truck and its suspension can be modeled, in first approximation, as a k - m - c spring-mass-damper system as shown in Figure 4.

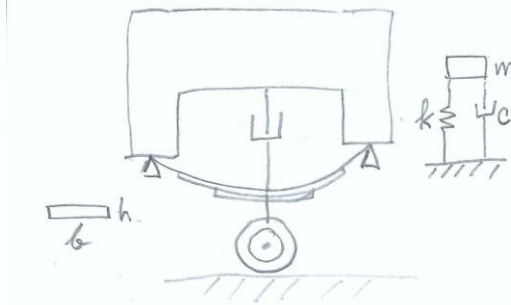


Figure 4 Truck suspension consisting of a leaf spring and an oil damper

The leaf spring can be approximated by a uniform beam of length $L=1200$ mm and cross-section dimensions $h=4$ mm, $b=40$ mm made of steel with $E=200$ GPa. The truck mass apportioned to the wheel can be estimated as $m=520$ kg. Assume initial displacement $u_0=105$ mm and zero initial velocity $\dot{u}_0 = 0$. Do the following:

- (1) Display input data
- (2) Calculate the second moment of area I of the beam cross-section
- (3) Calculate the flexural stiffness EI of the beam
- (4) Calculate the spring constant k of the system
- (5) Calculate the natural frequency in rad/sec ω_n and in Hz f_n
- (6) Calculate the critical damping c_{cr}
- (7) Assume 25% damping ratio, i.e., $\zeta=0.25$, and calculate the physical damping c . Print ζ and c
- (8) Calculate the damped frequency in rad/sec ω_d and in Hz f_d
- (9) Calculate the damped period of oscillation τ_d
- (10) Calculate the characteristic-equation roots s_1 , s_2 and the constants C_1 , C_2
- (11) Calculate the constants A , B and then the constants C , φ

- (12) Calculate the duration t_1 of one oscillation and the corresponding position $u_1=u(t_1)$
- (13) Calculate the value $u_{0.02}$ representing 2% of the initial displacement
- (14) Plot the response of the system for $N_{osc}=17$ oscillations and use datatips on the plot to find the following:
- (a) the largest response
 - (b) the response value calculated at item (12)
 - (a) the time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (13)
- (15) Discuss your results
- (16) **Graduate work:** assume the damper is worn out such that ζ is reduced to $\zeta=10\%$.
- (a) Repeat items (6) through (15)
 - (b) Compare and discuss the two results

B.5 OVERDAMPED SPRING-MASS-DAMPER SYSTEMS

Consider a spring-mass-damper system with natural frequency $f_n=5\text{Hz}$ and various values of the damping ratio ζ . The system is given an initial displacement $u_0 = 3 \text{ mm}$ with zero initial velocity.

Do the following:

- (1) display input data
- (2) calculate the rad/s frequency ω_n
- (3) Assume overdamped conditions $\zeta > 1$ and develop the solution as follows:
 - (a) write the expressions of the roots s_1, s_2 in terms of ω_n and ζ
 - (b) write the solution $u(t)$ in terms of s_1, s_2 and constants C_1, C_2
 - (c) write the expression of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$
 - (d) take $\zeta=2.5$ and calculate the numerical values of s_1, s_2 and C_1, C_2
 - (e) plot the response $u(t)$ for $t=0, \dots, 3$ seconds
- (4) Assume critically damped conditions $\zeta=1$ and do the following:
 - (a) write the expressions of the two partial solutions $u_1(t), u_2(t)$ in terms of ω_n
 - (b) write the expression of the complete solution in terms of ω_n and C_1, C_2
 - (c) write the expressions of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$
 - (d) calculate the numerical values of C_1, C_2
 - (e) plot the response $u(t)$ for $t=0, \dots, 3$ seconds
- (5) **Graduate Work:** nothing here
- (6) **Graduate Work:** some race-car manufacturers design for critical damping whereas others use overdamped design. Discuss the pros and cons of the two options

B.6 ANALYZE SPRING-MASS-DAMPER FORCED VIBRATION

NOTHING HERE

B.7 CALCULATE FLEX-BEAM RESPONSE TO SLIGHTLY UNBALANCED ELECTRIC MOTOR EXCITATION

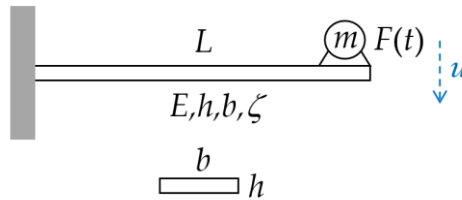


Figure 5 damped cantilever beam excited by force $F(t)$ from slightly unbalanced electric motor

Consider a cantilever beam of length $L=520$ mm, elastic modulus $E=200$ GPa, cross-section properties $h=4$ mm, $b=30$ mm and damping ratio $\zeta=15\%$. (The damping is achieved through a proprietary damping treatment applied on the beam surface.) An electric motor of mass $m=0.7$ kg operating at 3,100 RPM is attached at the beam tip as shown in Figure 5. The electric motor is slightly unbalanced producing a slight vibratory excitation $F(t) = \hat{F}e^{i\omega t}$ with $\hat{F} = 12$ mN. The excitation frequency ω depends on the electric motor RPM. Do the following:

- (1) Display input data
- (2) Calculate
 - (a) cross-section second moment of area I , flexural stiffness EI , and spring constant k
 - (b) natural frequency ω_n rad/s, critical damping c_{cr} , physical damping c
 - (c) quasi-static deflection u_{qst} corresponding to $\omega_n \rightarrow 0$
- (3) Calculate the values of
 - (a) resonant frequency ω_r and damped frequency ω_d
 - (b) peak magnification M_r
 - (c) magnification at phase resonance M_{90}
 - (d) physical response magnitude $|\hat{u}_{90}|$ at phase resonance
 - (e) calculate the Hz frequencies f_r, f_n and the corresponding RPM values

- (4) assume excitation at phase resonance:
- (a) calculate the excitation frequency ω in rad/s and f in Hz
 - (b) calculate period of oscillation τ
 - (c) Plot the time response for $N_{osc}=13$ oscillations. Hint: plot the real part of the complex displacement $u(t)$
- (5) **Graduate work:**
- (a) plot the magnitude and phase of the frequency response function FRF vs. excitation f Hz from quasi-static to 12 times the phase resonance frequency
 - (b) calculate the FRF magnitude at phase-resonance RPM and at double the phase resonance RPM
 - (c) calculate the vibration magnitude when the electric motor runs at its operational RPM
 - (d) comment on what vibration would be observed as the motor is started and accelerated to reach its operational RPM

**B.8 GRADUATE WORK: CALCULATE
HORIZONTAL FLEX-BEAM RESPONSE TO IMPACT**

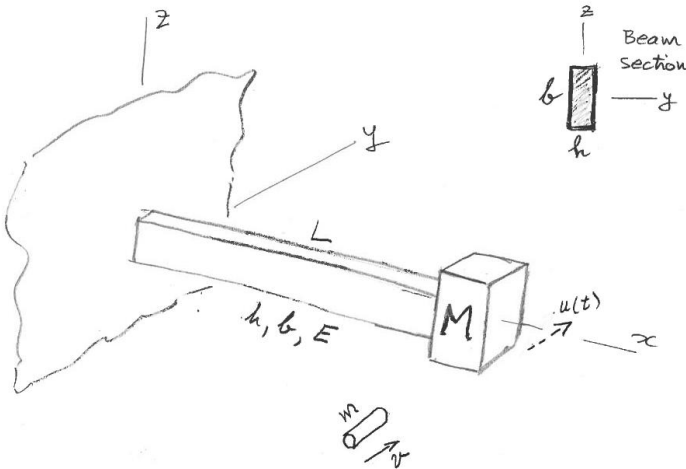


Figure 6: impact excitation of a horizontal flex-beam vibration system

Consider a cantilever beam of length $L=525$ mm with a mass $M=62\text{kg}$ attached at its tip as shown in Figure 6. The beam material is steel with $E=200$ GPa. The beam cross-section dimensions are $h=4$ mm, $b=30$ mm. The beam is initially at rest.

A small projectile of mass $m=13$ grams and velocity $v=1020$ m/s hits the mass and remains embedded in it. This impact imparts an initial velocity to the system that starts to oscillate with motion $u(t)$ contained in the horizontal plane. Do the following:

- (1) Display input data
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$
- (3) Calculate the cross-sectional stiffness EI of the beam
- (4) Calculate the flex-beam spring constant k
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass
- (6) Calculate the rad/sec natural frequency ω_n and period of oscillation τ
- (7) Calculate the position of the mass $u(t_1)$ after $t_1=17$ sec

- (8) Plot the response of the system for $N_{osc}=13$ oscillations and find on the plot:
- (a) the largest response
 - (b) the response calculated at item (7)
- (9) Discuss your results

C. Extra Credit Challenge Problems

Challenge problems are optional.

C.1 ANALYZE VERTICAL SPRING-MASS VIBRATION SYSTEM

NOTHING HERE

C.2 ANALYZE VERTICAL FLEX-BEAM VIBRATION

NOTHING HERE

C.3 ANALYZE HORIZONTAL FLEX-BEAM VIBRATION

NOTHING HERE

C.4 CALCULATE HORIZONTAL FLEX-BEAM RESPONSE TO INITIAL DISPLACEMENT

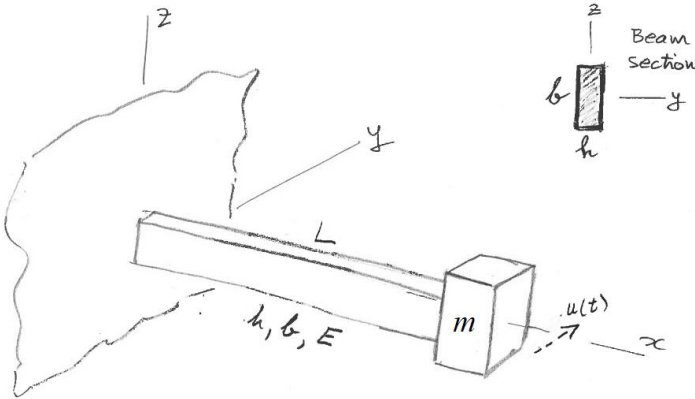


Figure 7: flex-beam vibrating system

Consider a cantilever beam of length $L=525$ mm with a mass $m=62\text{kg}$ attached at its tip as shown in Figure 7. The beam material is steel with $E=200$ GPa. The beam cross-section dimensions are $h=4$ mm, $b=30$ mm. The tip mass is given an initial displacement $u_0=3\text{mm}$ with zero initial velocity. Do the following:

- (1) Display input data
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$
- (3) Calculate the cross-sectional stiffness EI of the beam
- (4) Calculate the spring constant k of the system
- (5) Calculate the natural frequency in rad/sec ω_n
- (6) Calculate the natural frequency in Hz f_n
- (7) Calculate the period of oscillation in sec τ
- (8) Calculate the position of the mass $u(t_1)$ after $t_1=18$ sec
- (9) Plot the response of the system for $N_{osc}=13$ oscillations and use a data tip on the plot to find the value $u(t_1)$ at t_1 on the plot
- (10) Discuss your results

C.5 ANALYZE SPRING-MASS FORCED VIBRATION

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C.6 CALCULATE CANTILEVER RESPONSE TO SUDDENLY APPLIED WEIGHT

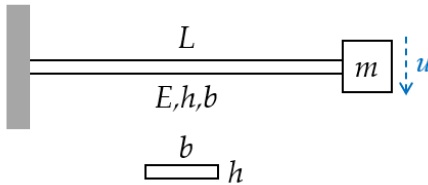


Figure 8: cantilever beam with tip mass

Take $g=9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a cantilever beam of length $L=525 \text{ mm}$ and a mass $m=3 \text{ kg}$ that has to be attached to its tip as shown in Figure 8. The beam is made of steel with elastic modulus $E=200 \text{ GPa}$ and yield stress $Y=710 \text{ MPa}$ cross-section properties $h=4 \text{ mm}$, $b=30 \text{ mm}$. Do the following:

- (1) Display input data
- (2) Calculate the cross-section second moment of area I , the flexural stiffness EI , the spring constant k , the natural frequency $\omega_n \text{ rad/s}$, $f_n \text{ Hz}$, and period of oscillation τ of the system
- (3) Calculate the weight F_0 of the mass m
- (4) Static case: assume the mass is attached onto the undeformed beam and the let go gradually without inducing any vibration and calculate the static displacement u_{st}
- (5) Dynamic case: assume the mass is attached onto the undeformed beam and then let go suddenly inducing a state of vibration:
 - (a) calculate the maximum dynamic displacement u_{dyn}
 - (b) Plot the response $u(t)$ measured from the undeformed position for $N_{osc}=13$ oscillations
 - (c) Put a datatip on the plot to identify the maximum displacement and compare with result of item (5)(a)
- (6) **Graduate work**: calculate the maximum stress at the root of the beam and the safety factor $SF=Y/\sigma$ and comment on your results for the above two cases:
 - (a) σ_{st} for the static case
 - (b) σ_{dyn} for the dynamic case
- (7) Discuss your results

C.7 ANALYZE FLEX-BEAM PENDULUM

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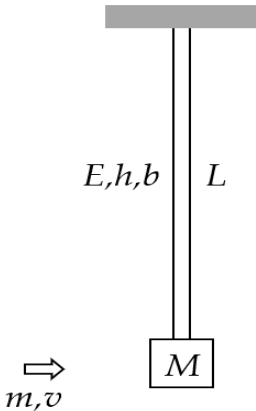
C.8 CALCULATE FLEX-BEAM PENDULUM RESPONSE TO IMPACT

Figure 9: impact excitation of a vertical flex-beam pendulum

Take $g=9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a vertical flex-beam pendulum of length $L=440 \text{ mm}$ with a mass $M=62 \text{ kg}$ attached at its end as shown in Figure 9. The beam material is steel with $E=200 \text{ GPa}$. The beam cross-section dimensions are $h=4 \text{ mm}$, $b=30 \text{ mm}$. The beam is initially at rest.

A projectile of mass $m=11 \text{ grams}$ and velocity $v=1020 \text{ m/s}$ hits the mass and remains embedded into it. The azimuthal orientation of the projectile motion is such that the **beam flexes about its minor cross-sectional stiffness**. This impact imparts an initial velocity to the system that starts to oscillate. Do the following:

- (1) Display input data
- (2) Calculate the minor second moment of area I of the beam
- (3) Calculate the minor cross-sectional stiffness EI
- (4) Calculate the flex-beam spring constant k
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as resulting from the interaction between the projectile and the large mass
- (6) Calculate the rad/s natural frequency ω_{nFBP} and period of oscillation τ_{FBP} of the flex-beam pendulum with projectile embedded into it
- (7) Calculate the response $u(t_1)$ after $t_1=18 \text{ sec}$

- (8) Plot the response of the system for $N_{osc}=23$ oscillations and find on the plot:
- (a) the largest response
 - (b) the response value calculated at item (7)
- (9) Discuss your results in comparison with results of Problem A.3

**C.9 COMPLEX EXPONENTIALS AND TRIGONOMETRIC
FUNCTIONS**

NOTHING HERE